Hamiltonian cycles on bicubic planar maps: bounds on the growth rate

Meander numbers: growth rate and exponent. Paper E.

Carl Johan Ragnarsson

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Abstract

A bicubic planar map is a map on the sphere in which every vertex has degree three and the vertices can be colored black and white so that every edge joins two colors. Let z_n be the number of such maps with $2n$ vertices, each with a Hamiltonian cycle and a marked oriented edge on the cycle. The limit $\mu^2 = \lim z_n^{1/n}$ exists and is estimated numerically at 10.113; the best bounds stated in the literature are $8 \le \mu^2 \le 16$. We prove

$$10.0869 \;\le\; \mu^2 \;\le\; 10.1571.$$

The maps are encoded as walks of a machine with two stacks of parities, and both bounds are certificates for this machine, checked by computer in exact arithmetic. The upper bound is a positive weight on the states such that the successors of every state weigh at most 3.187011595 times as much as the state; its 536,805,378 inequalities were checked by three programs, one of them written separately for this paper. The lower bound is a stationary random rule for choosing moves that looks only at the top ten symbols of each stack. The machine is reversible, and this makes exact stationarity easy: any positive numbers define such a rule. Its entropy, at least 1.1556 per move, gives the lower bound. All code was written using Claude (Opus 5.5) or Codex (Sol 5.1) with active review and guidance throughout.

1 Introduction

1.1 The problem

A *planar map* is a connected graph drawn on the sphere without crossings, loops and multiple edges allowed, considered up to deformation. It is *cubic* if every vertex has degree three, and *bicubic* if moreover its vertices can be colored black and white so that every edge joins a black and a white vertex. A *Hamiltonian cycle* is a cycle through every vertex exactly once. We count triples (bicubic planar map with $2n$ vertices, Hamiltonian cycle, one edge of the cycle marked and oriented); the marked edge removes all symmetry. Their number is z_n:

$$z_1, z_2, \ldots \;=\; 2,\; 8,\; 40,\; 228,\; 1424,\; 9520,\; 67064,\; 492292,\; \ldots$$

This is sequence A116456 of the OEIS [6]. Concatenating the codes of two configurations (Section 2) gives $z_{m+n} \ge z_m z_n$, so the limit

$$\mu^2 \;=\; \lim_{n\to\infty} z_n^{1/n} \;=\; \sup_n z_n^{1/n}$$

exists by Fekete's lemma [3]. We write the growth rate as a square, as the physics literature does, because z_n counts maps with $2n$ vertices: μ is the growth per vertex. (In this paper μ is always this bicubic growth rate, not the meander growth rate of Papers A–D.)

The problem comes from physics. Guitter, Kristjansen and Nielsen [4] introduced it as a model of two-dimensional quantum gravity coupled to matter of central charge $c = -1$, and argued that $z_n \approx \text{const} \cdot \mu^{2n} n^{\gamma-2}$ with the irrational exponent $\gamma = -(1+\sqrt{13})/6$. The same model is the bicubic counterpart of meanders: both count pairs of non-crossing arch systems on the two sides of a line, and in both the physical prediction rests on a coupling of matter and gravity that has not been proved (for meanders see [8]). Di Francesco, Duplantier, Golinelli and Guitter [1] computed z_n for $n \le 28$ and estimated

$$\mu^2 \;=\; 10.113 \pm 0.001$$

(their equation (60)). This is a numerical estimate, not a bound; the authors write that the predictions remain conjectural.

1.2 Previous bounds and the main result

The only rigorous bounds we found are those of Guitter, Kristjansen and Nielsen [4], Section 5. In arch language a configuration is two systems of non-crossing arches on $2n$ points of a line, one above and one below, every arch joining an even and an odd point. Forgetting the parity condition gives at most $c_n c_{n+1}$ configurations, with c_n the Catalan numbers, hence $\mu^2 \le 16$. Conversely, a single arch system above the line automatically satisfies the parity condition, and flipping any subset of its n arches below the line gives $2^n c_n$ distinct configurations, hence $\mu^2 \ge 8$. Later papers [1], [2] study exponents numerically and state no bounds.

Theorem 1. *For every $n \ge 1$,*

$$z_n \;\le\; \Bigl(\frac{3187011595}{10^9}\Bigr)^{2n}, \qquad \textit{hence} \qquad \mu^2 \le 10.157042906664444025\ldots < 10.15705;$$

and

$$\mu^2 \;\ge\; 10.086918840819.$$

So μ^2 is known to within 0.071, and the estimate 10.113 lies 0.026 above the lower bound and 0.044 below the upper bound.

1.3 The methods in outline

Both bounds work on a machine with two stacks (Section 2). Read the vertices along the Hamiltonian cycle one at a time. Each vertex has one edge that is not on the cycle, a *chord*, drawn inside or outside the cycle. The open chords on each side form a stack, and since the map is bipartite, all that matters about an open chord is the parity of the position where it was opened. The configurations of size n are exactly the closed walks of $2n$ moves from the empty state.

The upper bound (Section 3) is the method of Paper B [7], Sections 3 and 4, in its simplest form: a positive weight on states that no step increases by more than a factor R in total gives $z_n \le R^{2n}$. The weight looks at the top twelve symbols of each stack and is stored as a table of 67 million integers.

The lower bound (Section 4) is the entropy method of Paper B, Sections 5 and 6. One constructs a random rule for the next move that looks only at the top symbols of the stacks and that is stationary: the stacks seen from the top look statistically the same at every time. If the rule makes on average e^h equally likely choices per move, there are about e^{ht} closed walks of t moves. For meanders, making the rule exactly stationary was the main difficulty and needed a large exact linear solve and many corrections. Here it is free. The two-stack machine is reversible: every move can be undone by a move. Intuitively, a rule that is in balance across every move, as much flow forward as backward, is automatically stationary, and such rules are given by arbitrary positive numbers on the states. The work that remains is to choose good numbers and to bound the entropy.

1.4 Status of the results

The encoding by two-stack words and the upper certificate were obtained earlier in this project; the encoding was checked in a separate independent reading, which also recounted $z_1, \dots, z_{10}$ from the definition and counted the maps for $n \le 4$ without the word model, and which confirmed an upper certificate of radius 7 on all states with at most 22 open chords. The radius-12 certificate of Theorem 1 was first checked by two programs written by one author; for this paper it was checked a third time by a program written separately from the mathematical statement alone (Section 5). The lower bound is new. Its theorem (Theorem L) adapts the entropy theorem of Paper B; the proof of the adaptation, the reversible construction (Proposition 7) and the closing lemma (Lemma 4) were checked in a separate independent reading. The lower certificate was checked by two separately written programs, one of which checks every stationarity equation directly without using Proposition 7, and a third time by a program written in that independent reading. Nothing has been reviewed outside the project, and no proof assistant was used. Programs, proofs and certificates are in the project repository [5]; Appendix A lists where each proof is.

2 Words and the two-stack machine

2.1 From maps to words

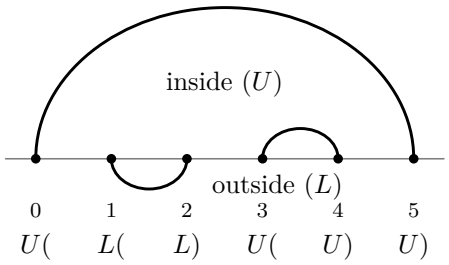
Take a configuration of size n and number the vertices $0, 1, \dots, 2n-1$ along the Hamiltonian cycle, starting at the head of the marked edge and following its orientation. Every vertex has degree three and two of its edges lie on the cycle, so exactly one edge at each vertex is a chord: an edge drawn inside the cycle or outside it. The cycle is a simple closed curve, so chords on the same side do not cross. Write the letter $U($ or $U)$ at a vertex whose chord lies inside, according to whether the vertex is the smaller or the larger end of its chord, and $L($, $L)$ for chords outside. The U-letters then form a word of matched parentheses (a Dyck word), and so do the L-letters.

Bipartiteness becomes a parity condition. Color each vertex by the parity of its number. The cycle edges join consecutive numbers, and $2n-1$ to 0, so they always join two parities; a connected bipartite map has only one coloring up to exchange of the colors, so the map is bipartite exactly when every chord joins an even and an odd position.

Theorem 2 (Encoding)**.** *z_n is the number of words of length $2n$ in the four letters $U(, U), L(, L)$ whose U-letters and L-letters each form a Dyck word, and in which every matched pair of*

parentheses sits at two positions of opposite parity.

The inverse map draws the U-pairs as chords inside a circle and the L-pairs outside. A map is determined by the cyclic order of the edges around each vertex, and here this order is read off the word, so the map is injective. The proof is in [5] (Appendix A); it was checked independently, and the counts were confirmed for $n \le 4$ by enumerating maps directly, without the word model. Figure 1 shows an example. Concatenating two words gives a word, because the second word is shifted by an even number of positions; so $z_{m+n} \ge z_m z_n$.



position	letter	U	L	c
0	$U($	0	–	1
1	$L($	0	1	0
2	$L)$	0	–	1
3	$U($	01	–	0
4	$U)$	0	–	1
5	$U)$	–	–	0

Figure 1: The word $U(\,L(\,L)\,U(\,U)\,U)$ of size 3. Positions are the vertices along the cycle. Every arch joins an even and an odd position. Right: the two stacks (bottom first, "–" for empty) and the parity after each letter.

2.2 The machine

A *state* is a triple (U, L, c) of two finite words over $\{0,1\}$, the *stacks*, written bottom first, and a parity $c \in \{0,1\}$. Intuitively, U lists the parities of the positions of the open chords inside, L those outside, and c is the parity of the next position. There are four moves; each flips c:

- pushU: append c to U; pushL: append c to L (always allowed);
- popU: remove the top symbol of U, allowed if U is nonempty and its top is $1-c$; popL likewise.

Reading the letters $U(, L(, U), L)$ as the four moves, a word of Theorem 2 is the same as a walk of $2n$ moves from the empty state $e = (\varnothing, \varnothing, 0)$ back to e: a closing parenthesis must match the most recent open one, which is the top of the stack, and the parity condition says that its parity differs from the current one. So $z_n = (A^{2n})_{e,e}$ for the transfer matrix A of the machine.

Three facts about the machine are used below.

Lemma 3. (i) Symmetry. *y is a successor of x if and only if x is a successor of y; so A is symmetric, and a walk read backward is a walk.* (ii) Charge. *Let N_0, N_1 be the numbers of zeros and ones in the two stacks together. Every state reached from e has $N_0 - N_1 = c$.*

Proof. (i) Pushing c at parity c leads to parity $1-c$ with top $c \ne 1-c$, so the pop is allowed and undoes the push; conversely a pop of the top $1-c$ at parity c is undone by pushing $1-c$ at parity $1-c$. (ii) Pushing $c=0$ raises $N_0 - N_1$ by one and changes c from 0 to 1; pushing $c=1$ lowers it by one and changes c from 1 to 0; pops are the reverse. □

The third fact is that every state allowed by the charge rule can be emptied quickly. It replaces the sweep lemma of Paper B.

Lemma 4 (Sweep)**.** *Every state (U, L, c) with $N_0 - N_1 = c$ is joined to e, in both directions, by walks of at most $5|U| + 3|L|$ moves.*

Proof. By symmetry it suffices to go from the state to e. A *transfer* from U to L, when the top of U is c, pushes c on L and then pops U: two moves that move the top symbol across and keep the parity. At parity c repeat the first rule that applies: (1) if the top of U is $1-c$, pop U; (2) if the top of L is $1-c$, pop L; (3) if U contains a symbol $1-c$, transfer from U to L; (4) otherwise transfer from L to U. In (4) the stack L is nonempty unless the state is e: otherwise $U = c^k$ and the charge rule forces $k = 0$, $c = 0$. When (4) first applies, U holds a single symbol repeated; from then on U only receives that symbol, and whenever rules (1) and (2) fail the parity equals it, so (3) never applies again. Hence each symbol of U is transferred at most once by (3), each symbol at most once by (4), and each is popped once: at most $|U| + |L|$ pops and $2|U| + |L|$ transfers. □

Intuitively, the two stacks act as one tape with a cursor between them, and a transfer moves the cursor across a symbol equal to the current parity. A program checked Lemma 3 and Lemma 4 on all states with at most 10 symbols; the worst case there uses 2.6 moves per symbol.

3 The upper bound

3.1 A weight that the moves do not increase

Lemma 5. *Let $v > 0$ be a function on all states and $R > 0$, with*

$$\sum_{y \text{ successor of } x} v(y) \;\le\; R\, v(x) \qquad \text{for every state } x.$$

Then $z_n \le R^{2n}$ for every n.

Proof. By induction, the total weight of the endpoints of all walks of t moves from x is at most $R^t v(x)$. The closed walks at e contribute $(A^{2n})_{e,e}\, v(e)$ to this total for $t = 2n$; divide by $v(e)$. □

The bound holds for every n with constant one, with no asymptotic input. The work is to find v and to check infinitely many inequalities with finitely many computations.

3.2 The weight

Fix a *radius* $r \ge 2$. For a stack S let $E(S)$ be the number of adjacent equal pairs (positions i with $S_i = S_{i+1}$) and $T(S)$ the number of adjacent equal triples. Let view(S) be the top r symbols of S, or all of S together with a bottom mark if S is shorter. The weight is

$$v(U, L, c) \;=\; a^{|U|+|L|}\, p^{E(U)+E(L)}\, t^{T(U)+T(L)}\, u\big(\text{view}(U), \text{view}(L), c\big),$$

with three positive numbers a, p, t and a table u of positive integers indexed by the pairs of views and the parity. Intuitively, a prices the number of open chords, p and t price runs of equal parities, which are rare in typical stacks, and the table corrects the price for the top of each stack, where the moves act.

The inequality of Lemma 5 at a state depends only on its top $r + 1$ symbols in each stack.

Lemma 6 (Truncation)**.** *Let x' be obtained from x by keeping the top $r+1$ symbols of each stack. Then $\sum_y v(y)/v(x) = \sum_{y'} v(y')/v(x')$.*

Proof. The allowed moves agree, because they depend only on the top symbols. For each move the quotient $v(y)/v(x)$ is $a^{\pm 1}$ times p and t to the changes of E and T, times $u(y)/u(x)$. A push or pop changes E and T only through the top three symbols, and the views before and after the move are read from the top $r+1$ symbols; these agree for x and x' since $r+1 \geq 3$. □

So the infinitely many inequalities reduce to the states with at most $r+1$ symbols in each stack: $2(2^{r+2}-1)^2$ of them, which is 536,805,378 at $r = 12$. In the language of [7] these are the rows of the certificate; here each row is simply a short state.

3.3 The certificate

The certificate has radius 12, parameters

$$a = \tfrac{1783}{2000}, \qquad p = \tfrac{137}{200}, \qquad t = \tfrac{97}{125}, \qquad R = \tfrac{3187011595}{10^9},$$

and a table of 67,092,481 used entries (the entries for parity 1 are those for parity 0 with the two parities exchanged, which is only a way of storing the table). All inequalities hold; with R lowered by 10^{-9} four of them fail, so R is the least admissible value with this denominator. The largest quotient $\sum_y v(y)/v(x)$ is 3.187011594348, at $U = 0$, $L = 1111000000$ (bottom first), $c = 0$. By Lemma 5, $z_n \leq R^{2n}$, which is the upper half of Theorem 1.

The table was found by an iterative solver in floating point; the solver plays no role in the proof. Smaller radii give weaker bounds: $R^2 = 10.187205938372711824$ at radius 7, and in floating point 10.2646, 10.2323, 10.2135, 10.1978 at radii 3 to 6. Without the factors p and t the bound at radius 8 is 10.2963, worse than radius 3 with them. The gap to the estimate 10.113 shrinks roughly like r^{-1}; a gap of 0.01 would need a radius of about 50, far beyond tables of size 4^{r+1}.

4 The lower bound

4.1 Stationary random rules

For a lower bound we must exhibit many closed walks. We let the machine run on infinitely deep stacks and choose each move at random by a rule that looks only at the top of the stacks.

Infinite stacks and windows. Let Ω be the set of triples $\omega = (U, L, c)$ with $U = (U_1, U_2, \ldots)$ and $L = (L_1, L_2, \ldots)$ infinite sequences over $\{0, 1\}$, indexed from the top, and c a parity. The four moves act on Ω as before; a pop needs top symbol $1 - c$. Fix a radius $r \geq 1$. The *window* of ω is $w = (U_1 \ldots U_r,\ L_1 \ldots L_r,\ c)$.

The reference law. Below the window the stacks are filled by a fixed background law β: a stationary Markov chain on $\{0, 1\}$, read downward, of memory $m \leq r$, with all transition probabilities positive and *reversible*, that is, its law is unchanged when finite blocks are read backward. We use the simplest case: the next symbol differs from the previous one with probability s and equals it with probability $1 - s$, with $0 < s < 1$ (a two-state chain is always reversible). For a word $w_1 \ldots w_k$ with $k \geq m$, its *continuation* is the conditional law of the following symbols given $w_1 \ldots w_k$.

Flows. A *flow* is a family of numbers $J(w,\xi) \geq 0$ with total 1, one for each window w and each move ξ allowed at w. Put $\nu(w) = \sum_\xi J(w,\xi)$ and $p(\xi \mid w) = J(w,\xi)/\nu(w)$. The flow defines a random infinite state: the window has law ν, and the two stacks continue below it as independent continuations under β. It also defines a random move, chosen with probability $p(\xi \mid w)$. Intuitively, $J(w,\xi)$ is the probability of seeing window w and then making move ξ. The conditions are:

(S) *Stationarity.* For every *extended window* Y, with $r+1$ symbols of each stack, the probability that the extended window after one move is Y equals its probability now, which is $\nu(Y^\circ)$ times the reference probabilities of the two symbols at depth $r+1$ given the symbols above them; Y° is Y without those two symbols.

(D) *Zero drift.* The total flow of pushU equals that of popU, and the same for L.

(P) *Positivity.* $J(w,\mathsf{pushU}) > 0$ and $J(w,\mathsf{pushL}) > 0$ for every window w.

Theorem L. If a flow satisfies (S), (D) and (P), then $\mu \geq e^h$, that is, $\mu^2 \geq e^{2h}$, where

$$h \;=\; \sum_{w,\xi} J(w,\xi)\, \log \frac{\nu(w)}{J(w,\xi)}$$

is the entropy of one move given the window.

The proof follows that of Theorem F of Paper B [7] in three steps.

Step 1: stationarity. After a push the symbols below the new extended window are old symbols that continued the old window; after a pop the two newly revealed symbols are fresh continuations. So, given the new extended window, the rest is again a continuation (this uses $m \leq r$), and by (S) the law of the new extended window is the old one. Hence the random infinite state has the same law after every move.

Step 2: ergodicity. Let f be a bounded function with $\mathbb{E}[f(\omega_1) \mid \omega_0] = f(\omega_0)$. Stationarity gives $\mathbb{E}(f(\omega_1) - f(\omega_0))^2 = 0$, so f is unchanged by every move of positive probability, in particular by both pushes, by (P). Comparing the two pushes from parity 1 gives $f(1U, L, 0) = f(U, 1L, 0)$. For the other symbol, use the three comparisons

$$f(0U, L, 0) = f(0U, 0L, 1) = f(U, 00L, 1) = f(U, 0L, 0).$$

The first and last compare a push on L; the middle compares the two pushes from $(U, 0L, 0)$. Thus $f(xU, L, 0) = f(U, xL, 0)$ for either symbol: moving a top symbol between stacks does not change f. Glue the two stacks into one two-sided sequence $(\ldots, U_2, U_1 \mid L_1, L_2, \ldots)$; then f is invariant under shifting this sequence. Because β is reversible with positive transitions, the glued sequence has the same null sets as the stationary two-sided chain, which is ergodic under the shift. So f is constant almost surely, and the process is ergodic.

Step 3: counting. By the ergodic theorem and (D), in t moves both stack heights change by $o(t)$, and the product of the probabilities of the chosen moves is about e^{-ht}. So from some fixed infinite state there are at least $\frac{1}{2}e^{t(h-\varepsilon)}$ move sequences of length t along which no stack ever drops by more than εt. They only read the top $\varepsilon t + 1$ symbols of each stack. Copy those symbols into a finite state, adding below them a run of equal symbols so that the charge rule of Lemma 3 holds. By Lemma 4 this state is reached from e, and every end state returns to e, in $O(\varepsilon t)$ moves. This gives $e^{t(h-\varepsilon)}/O(t)$ distinct closed walks of one length between t and $t + O(\varepsilon t)$, and $z_n \leq \mu^{2n}$ gives $h \leq \log \mu$ as $t \to \infty$ and $\varepsilon \to 0$. □

4.2 Reversible flows

For meanders, the hard part was to make (S) hold exactly: it is a large linear system, and Paper B solved it at a small radius and corrected the solution with about a million closed-walk directions. For the two-stack machine there is a direct construction. Write $\beta(a)$ for the stationary probability of a word a (read from the top), $\beta(w) = \beta(U_1 \dots U_r)\,\beta(L_1 \dots L_r)/2$ for a window w, and $q(x \mid w)$ for the probability that the symbol above the top of U is x given the window; for our chain $q(x \mid w)$ is s if $x \neq U_1$ and $1 - s$ otherwise. Let w^U be the window after pushU at w.

Proposition 7. *Let $X_U(w) > 0$ be given for every window with $U_1 \neq c$, and $X_L(w) > 0$ for every window with $L_1 \neq c$. Put*

$$J(w, \mathsf{popU}) = \beta(w)\, X_U(w), \qquad J(w, \mathsf{pushU}) = \beta(w)\, q(c \mid w)\, X_U(w^U),$$

and the same for L. After normalization, J satisfies (S), (D) and (P).

Proof. The random infinite state of the flow has density $\nu(w)/\beta(w)$ with respect to the product of two stationary chains and a fair parity. The move pushU maps ω to ω' with one more symbol, c, on top of U, and the reference probability is multiplied by $q(c \mid w)$, because for a reversible chain the probability of the symbol above equals that of the symbol below. The flow forward along this move is $J(w, \mathsf{pushU})/\beta(w)$ per unit of reference probability at ω, that is $q(c \mid w)X_U(w^U)$; the flow back along popU from ω' is $J(w^U, \mathsf{popU})/\beta(w^U) = X_U(w^U)$ per unit of reference probability at ω', and the reference probability at ω' is $q(c \mid w)$ times that at ω. So the two flows are equal. Every move is undone by a move, so at every infinite state the total inflow equals the total outflow, and (S) follows by summing over the states with a given extended window. Summing the equality over all pushes on U gives (D). (P) holds since every $X > 0$. □

Intuitively, the rule is in detailed balance: in the long run it crosses every move as often forward as backward. Such a rule is automatically stationary and automatically has zero drift. Any positive numbers X give a valid lower bound; optimization only chooses them well. At an optimum, $\log X(v)$ is a weighted average of $\log(\nu/J)$ over the flow values that use $X(v)$, and iterating this relation converges quickly.

4.3 The certificate

We take $s = 41/50$: the reference stacks prefer to alternate, as typical stacks do. With $s = 1/2$ (independent symbols) the bounds are smaller, 9.983 instead of 10.084 at radius 9. The numbers X were scaled so that the largest is 2^{60} and rounded to integers; then every flow value is an integer after a common scaling, and only the entropy needs rounding. Table 1 gives the bound by radius.

At radius 10 the certificate has 2,097,152 windows and 6,291,456 flow values, and

$$h \;\geq\; 1.15561971006089, \qquad \mu^2 \;\geq\; e^{2h} \;\geq\; 10.086918840819,$$

the lower half of Theorem 1. Already radius 1, a rule that sees only the top symbol of each stack, gives about 9.497 in floating point, well above 8. The bound grows slowly with the radius, by 0.0076 from radius 8 to 10, so the gap of 0.026 to the estimate would need much larger windows.

radius r	windows	e^{2h}	status
1	8	9.497	float
2	32	9.880	float
4	512	10.030980878326	certified
6	8,192	10.065	float
8	131,072	10.079276064125	certified
10	2,097,152	**10.086918840819**	certified

Table 1: Lower bounds $\mu^2 \geq e^{2h}$ from reversible flows with $s = 41/50$. Certified values are rounded down and were accepted by two checkers and by a third written in an independent reading; float values are from the optimizer and are not bounds.

5 How the certificates were checked

5.1 The upper certificate

The radius-12 certificate is a file of 268 MB (SHA-256 beginning `3e71e227`). Three programs accept it. Two were written with the solver and evaluate the 536,805,378 inequalities from formulas for the changes of E and T, in 128-bit integers. The third was written for this paper from the mathematical statement only, without reading the other two. It does not use those formulas: it enumerates the short states of Lemma 6 directly, applies the four moves literally, recounts E and T on the whole stacks before and after each move, reads the views off the stacks, and compares both sides after clearing denominators, in unsigned 128-bit integers; a bound computed at start-up shows that no intermediate value exceeds 2^{127}. It finds no violation, finds four violations when R is lowered by 10^{-9}, and locates the same largest quotient as the other two programs. It runs in 5 seconds on four threads.

A certificate that is wrong in a small way must be rejected. The third program rejects eight damaged copies: R lowered by 10^{-9}; the entry of the tightest state lowered by 0.1 percent; the entry of the tightest state with two full views of twelve symbols lowered by 0.1 percent; an entry used by a successor of the tightest state raised by 0.1 percent; a zero entry; a truncated file; a trailing byte; a wrong header. The full-view case matters: an earlier test on complete states could not reach it, because it needs 24 open chords. At radius 7 the same program reproduces the result of an independent check on all states with at most 22 open chords (largest quotient 3.191740267742 at U empty, $L = 000000$). The per-size inequality $z_n \leq R^{2n}$ also holds for the 34 published values, as it must.

5.2 The lower certificate

The radius-10 certificate is a file of 16 MB listing the integers X. Two programs, written separately, accept it. Checker A builds the flow from Proposition 7 in exact integers, checks positivity and that both drift sums vanish exactly, and encloses h in 160-bit interval arithmetic. Checker B does not use Proposition 7: it rebuilds the flow with its own reader and checks every balance equation (S) directly, 8,388,608 of them, in exact integers; then it checks (D) and (P) and bounds h from below with correctly rounded decimal logarithms, adding or subtracting one unit in the last place by hand and rounding every sum toward the safe side. Both give $e^{2h} \geq 10.086918840819$. As a test of checker B, three wrong constructions are rejected because

(S) fails: pushes without the reference factor q, one flow value changed by one unit, and a pop that reveals one fresh symbol instead of two.

A third program was written in the independent reading of the lower bound, from the statement of Theorem L and the file format alone. It checks (P) and (D), checks every equation (S) by sending each window, move and revealed symbol to its new extended window in exact integers, and bounds the logarithms with its own directed rounding. It confirms $e^{2h} \geq 10.086918840819$ at radius 10, with slack about $6 \cdot 10^{-13}$, reproduces the radius-4 and radius-8 values, and rejects damaged flows and the target 10.086918840820.

5.3 Limits

Both methods converge slowly in the radius. The upper gap to the estimate is about $0.49/r$, the lower gap about 0.026 at radius 10; it shrank by 0.014 from radius 6 to 8 and by 0.0076 from 8 to 10. A radius-12 lower certificate would have 33.5 million windows and needs a compiled checker; we expect a gain of a few thousandths. The two bounds use the same machine and the same notion of a window, but not the same information: the upper weight prices runs of equal symbols explicitly, while the lower rule uses a reference law that only prefers alternation. A reference law of memory two or three, or a lower rule that sees more of one stack than of the other, may close the gap faster. We have not tried.

A Where the proofs are

Paths are relative to the directory `meandric/research` of the project repository [5]. "Read independently" means that a separate reading, in the folder named, wrote its own proof or its own checking program. Nothing has been reviewed outside the project. The upper certificate (268 MB) is kept on external storage and referenced from its folder by name and SHA-256 hash.

Three numbers to reproduce first with a new implementation: the transfer gives

$$z_1, \ldots, z_{10} \;=\; 2,\ 8,\ 40,\ 228,\ 1424,\ 9520,\ 67064,\ 492292,\ 3735112,\ 29114128;$$

the upper certificate gives the largest quotient 3.187011594348; the radius-4 lower certificate gives $e^{2h} \geq 10.030980878326$.

result	where	status
Theorem 2 (encoding), transfer, $z_{m+n} \geq z_m z_n$	`602_bicubic_upper_bound/PROOF.md`, sections 2–3 (Theorems A, B, Lemma 2.1)	proved; read independently (`602r_referee_bicubic_bound`); counts recomputed, maps counted for $n \leq 4$
Lemma 3 (symmetry, charge), Lemma 4 (sweep)	`1367_bicubic_note/PROOF_LOWER.md`, section 1 (Lemmas 1, 2, S); test `sweep_test.py`	proved; tested on all states with at most 10 symbols; read independently (`1377_referee_bicubic_lower`)
Lemma 5, the weight, the row form	`602_bicubic_upper_bound/PROOF.md`, sections 4–5 (Lemma 4.1, covering lemma, Theorem C)	proved; read independently (`602r`)
Lemma 6 (truncation)	`1367_bicubic_note/PROOF_CHECK.md` (Lemma T)	proved; equivalent to the covering lemma of `602`
Upper certificate, radius 12	`602_bicubic_upper_bound/cert/cert_r12.bin`, `logs/cert_r12.log`; third check `1367_bicubic_note/check_states.c`, `logs/check_r12.log`, `logs/teeth_r12.log`	certified: three checkers, two by one author; radius 7 confirmed independently (`602r`)
Theorem L (entropy bound)	`1367_bicubic_note/PROOF_LOWER.md`, section 2; adapted from `805_entropy_certificates/PROOF.md` and Paper B, Section 5	proved; read independently (`1377`)
Proposition 7 (reversible flows)	`1367_bicubic_note/PROOF_LOWER.md`, section 3	proved; read independently (`1377`); checked by checker B
Lower certificates, radius 4, 8, 10	`1367_bicubic_note/cert/flow_r10.bin` (and `_r4`, `_r8`); `check_flow_a.py`, `check_flow_b.py`; `logs/check_flow_*`; third check `1377_referee_bicubic_lower/check_flow_da.py`	certified: three checkers, one written independently
Float tables	`602_bicubic_upper_bound/REPORT.md`, section 4; `1367_bicubic_note/logs/float_table.log`	numerical, not bounds
Literature	`1367_bicubic_note/FINAL.md` (primary sources); `602_bicubic_upper_bound/REPORT.md`, section 1; `189_bicubic_hamiltonian/REPORT.md`	literature

Format of the lower certificates. A file `flow_r*.bin` starts with the eight bytes `BICFLOW1`, then a 32-bit integer r (the radius), then 64-bit integers s_n, s_d (the switch probability $s = s_n/s_d$ of the reference law), $n_A = 4^r$ and $n_B = 4^r$, followed by $n_A + n_B$ positive 64-bit integers X (all little-endian). Write $\mathrm{code}(w_1 \ldots w_k) = \sum_i w_i 2^{i-1}$, with the top symbol in the lowest bit, and let c be the parity of the window. The number $X_U(w)$, for windows with $U_1 = 1 - c$, is entry $c\,2^{2r-1} + \mathrm{code}(L_1 \ldots L_r)\,2^{r-1} + \mathrm{code}(U_2 \ldots U_r)$ of the first block, and $X_L(w)$, for windows with $L_1 = 1 - c$, is the same entry of the second block with $\mathrm{code}(L_2 \ldots L_r)\,2^r + \mathrm{code}(U_1 \ldots U_r)$

in place of the middle terms. By Proposition 7 any positive X gives a valid flow, so a wrong reading of the format could only lower the bound, never invalidate it.

References

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[5] Meandric numbers project. Proofs, programs and data. `https://github.com/carljohanr/meandric-public`, 2026. Selected research release. Historical folder references in the appendices include material outside this subset; see the repository's reproduction guide.

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[7] Carl Johan Ragnarsson. The tape machine and bounds on the growth rate of meanders. Paper B of this series, draft, 2026.

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